

Mixed-replica polarization, diagonal heat–sextic comparison, Gaussian-ray repair, and shell boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-131 left two apparently separate problems. First, diagonal sample Grams do not determine the mixed conditional response. Second, a fixed heat law does not by itself give a state-uniform Ξ gap along large singlet rays, although the once-owned terminal sextic grows strongly there. R-132 resolves the exact algebraic relation between these issues and identifies one production-relevant positive model.

1. The complete square-of-conditional-mean term and trace term admit an exact two-future-replica polarization, provided the heat variable is kept common and same-root visits are recombined first.
2. The mixed six-row contraction has an exact Pauli–Fierz formula. Every floor denominator is localized to the two rational scalars τ, τ' , so the polynomial and rational shell problems can be distinguished without splitting the owner.
3. The diagonal heat Gram plus the once-owned sextic is globally coercive. For the exact 64-atom product-Rademacher comparison law, the raw diagonal form has the explicit constant $9/10240$. A half-normalized comparison still has the exact constant $1/2250$.
4. That diagonal theorem cannot be promoted to the physical mixed response: at the origin the same symmetric law has $\mathbb{E}\Xi = 0$ and zero sextic Hessian. This is an exact square-before-average boundary, not a numerical shortfall.
5. An exact two-point rational-ray law makes the mixed current–trace Hessian diverge like $-10c_1/(81\delta)$. Hence no law-free floor-uniform theorem is possible from the six-row algebra alone.
6. In contrast, a standard-Gaussian rational-ray packet has a floor-uniform score-transfer bound. The source and terminal sextic, each used once, leave an explicit positive margin exceeding $3/4$ on that entire scalar ray. This proves that production-law information can repair the two-point obstruction and selects the next route.

The result is T4. It proves no owner-complete production response, no cutoff-, chart-, control-, filtration-, visit-, output-, or refinement-uniform C_{mix} , C_{far} , or c_{bal} , no complete low block, matching energy, strict shifted-Douglas gap, absolute anchor, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, removal theorem, interacting measure, or Sector-A closure.

2. One-use owner and conditioning firewall

Fix a finite cutoff, positive floor, one temporally faithful chart, and one conditional law. Condition on the strict past and retained root. Recombine all visits of that root before polarization, and let the orthogonal

output clusters satisfy $\sum_{\mathcal{C}} P_{\mathcal{C}} = I$. The heat variable r is common to the two future replicas. It remains an $L^2(\mu_{\Sigma})$ coordinate and is neither independently replicated nor averaged before squaring.

The accepted coordinate equalities are

$$\begin{aligned} q_{\text{comp}} &= D^2 \sum_{k, \mathcal{C}} \mathbb{E} \mathcal{P}_{k, \mathcal{C}}^{\text{comp}} = -\frac{1}{2} D^2 \sum_k \mathbb{E} \mathcal{T}_k \\ &= \frac{1}{2} D^2 \sum_k \mathbb{E} \{ \mathcal{F}_{063, k}^{\text{ad}} - \mathcal{V}_{\text{fut}, k} \}. \end{aligned} \quad (2.1)$$

The R-063 forest and future variance in the last line are alternate coordinates for the same primitive trace owner, not additional reserves. The conditional and complete low pieces occur once each. The rational $\mathcal{R}_Q, \mathcal{M}_U, \mathcal{K}_R$ pieces occur once. The regular Cartan FAR square and the live rational curl 2680/729 are distinct owners and occur once. Heat stays internal. Source and terminal sextic are appended once only after the physical response is formed.

Consequently the fixed-chart action Hessian has the form

$$D^2 \mathcal{A}_J(h)[u, v] = \frac{9}{10} \langle u, v \rangle + q_{\text{comp}}(L_{\text{pred}} u, L_{\text{pred}} v) + q_6(Z; L_{\text{pred}} u, L_{\text{pred}} v). \quad (2.2)$$

The source term remains outside the physical pullback. None of the comparisons below moves it inside or spends it twice.

3. Exact mixed-replica response representation

For owner k , cluster $\mathcal{C}$, and spatial direction i , put

$$X_f = P_{\mathcal{C}} C_6(W_{f, r}) \partial_i W_{f, r}, \quad Y_{f, g} = P_{\mathcal{C}} C_6(W_{f, r}) \partial_i G'_{k, g}, \quad (3.1)$$

where f, f' are conditionally independent copies of the complete future, and G'_k is the independent Gaussian trace probe with the prescribed covariance. For fixed-law physical variations z, w , define

$$\begin{aligned} X_{f; z} &= P_{\mathcal{C}} \{ DC_6(W_f)[z_f] \partial_i W_f + C_6(W_f) \partial_i z_f \}, \\ X_{f; zw} &= P_{\mathcal{C}} \{ D^2 C_6(W_f)[z_f, w_f] \partial_i W_f + DC_6(W_f)[z_f] \partial_i w_f + DC_6(W_f)[w_f] \partial_i z_f \}, \end{aligned} \quad (3.2)$$

and define $Y_{f, g; z}, Y_{f, g; zw}$ by the same coefficient derivatives with the trace probe held fixed.

Theorem 3.1 (paired-replica polarization). Under the firewall above,

$$\begin{aligned} q_{\text{comp}}(z, w) &= \sum_{k, \mathcal{C}, i} \mathbb{E}_r \mathbb{E} [\mathbb{E}_{f, f'} \text{Re} \{ \langle X_{f; z}, X_{f'; w} \rangle + \langle X_f, X_{f'; zw} \rangle \} \\ &\quad - \mathbb{E}_{f, g} \text{Re} \{ \langle Y_{f, g; z}, Y_{f, g; w} \rangle + \langle Y_{f, g}, Y_{f, g; zw} \rangle \}]. \end{aligned} \quad (3.3)$$

Indeed,

$$\| \mathbb{E}_f X_f \|^2 = \mathbb{E}_{f, f'} \text{Re} \langle X_f, X_{f'} \rangle, \quad (3.4)$$

and differentiating one half of (3.4) produces four product-rule terms. Replica exchange pairs them into the two terms displayed in (3.3). The trace probe has the identical calculation. Thus (3.3) is an exact representation of the R-131 physical form. It is not a lower bound: the mixed contractions may have either sign and require the full joint law.

4. Mixed Pauli–Fierz algebra

For the doublet Pauli matrices σ_A , put

$$m_A(u) = u^\dagger \sigma_A u, \quad j_A(u, v) = 2 \operatorname{Re}(u^\dagger \sigma_A v), \quad (4.1)$$

$$\tau = \frac{2 \operatorname{Re}(u^\dagger v + \bar{\chi} w)}{|u|^2 + |\chi|^2 + e}, \quad \ell_A = j_A - \alpha \tau m_A, \quad \alpha = \frac{5}{9}. \quad (4.2)$$

For two replicas define JJ, JM, MJ, MM by summing the corresponding Pauli products. Completeness gives

$$\langle C_6(u)v, C_6(u')v' \rangle = c_0 \text{JJ} + c_1 \{ \text{JJ} - \alpha \tau' \text{JM} - \alpha \tau \text{MJ} + \alpha^2 \tau \tau' \text{MM} \}. \quad (4.3)$$

The invariant contractions are

$$\begin{aligned} \text{MM} &= 2|\langle u, u' \rangle|^2 - |u|^2 |u'|^2, \\ \text{JM} &= 4 \operatorname{Re}\{\langle v, u' \rangle \langle u', u \rangle\} - 2|u'|^2 \operatorname{Re}\langle u, v \rangle, \\ \text{MJ} &= \text{JM with primed and unprimed variables exchanged}, \\ \text{JJ} &= 2 \operatorname{Re}\{2\langle u, v' \rangle \langle u', v \rangle - \langle u, v \rangle \langle u', v' \rangle + 2\langle u, u' \rangle \langle v', v \rangle - \langle u, v \rangle \langle v', u' \rangle\}. \end{aligned} \quad (4.4)$$

On the diagonal these reduce to the R-130 radial-plus-wedge norm. Formula (4.3) is the cancellation obtained only after all Pauli rows are summed. Every floor denominator is confined to τ, τ' , but it has not disappeared.

If the four raw polynomial inputs are Q_r -supported, the audited raw JJ, JM, MJ, MM products are supported in Q_{r+2} , so the corresponding raw projection vanishes for $m \geq r + 4$. This is a localization candidate. Common heat, replica conditioning, trace, forest, and all visits still have to be intertwined before it becomes a production stopped-response theorem.

5. Diagonal heat–sextic comparison

Write $z = (u, \chi)$, $V = (v, w)$, $\rho = |u|^2 + |\chi|^2$, and

$$\frac{1}{4} \|\Xi(z, V)\|^2 = c_0 a^2 + c_1 \{(1 - \lambda)a - \lambda s\}^2 + (c_0 + c_1) |h|^2, \quad (5.1)$$

where $a = \operatorname{Re}(u^\dagger v)$, $s = \operatorname{Re}(\bar{\chi} w)$, $h = u_1 v_2 - u_2 v_1$, and $\lambda = \alpha |u|^2 / (\rho + e)$. The once-owned terminal sextic is $F_6 = 3\rho^3/20$, with exact Hessian

$$Q_6(z)[V, V] = \frac{9}{10} \rho^2 \|V\|^2 + \frac{18}{5} \rho \{\operatorname{Re}\langle z, V \rangle\}^2. \quad (5.2)$$

For a full-support heat law μ with finite second moment, define the diagonal comparison form

$$G_\mu(z)[V, V] = \int \|\Xi(z + A, V)\|^2 \mu(dA). \quad (5.3)$$

If (5.3) vanishes, continuity and full support imply $\Xi(z + A, V) = 0$ for every A . Varying the doublet in the first row gives $v = 0$. With $v = 0$, varying a nonzero doublet and then the singlet in the second row gives $w = 0$. Thus $G_\mu(z)$ is positive definite pointwise. Continuity gives a positive minimum $m_{\mu, R}$ on every compact ball $\rho \leq R$, while (5.2) gives $9R^2/10$ outside. Hence

$$G_\mu(z) + Q_6(z) \succeq \min\{m_{\mu, R}, 9R^2/10\} I. \quad (5.4)$$

This is a diagonal theorem, not yet the mixed response theorem (3.3).

The exact 64-atom comparison

Let every one of the six real heat coordinates be an independent sign. On $\rho \leq R$, set

$$D_R = (\sqrt{R} + \sqrt{6})^2 + e. \quad (5.5)$$

The wedge moment is

$$\mathbb{E}|h_A|^2 = |h|^2 + 2|v|^2. \quad (5.6)$$

Using the wedge row for v leaves the complete radial pair for the Schur bound. Since

$$\frac{c_0 c_1}{c_0 + c_1} = \frac{243}{28250P}, \quad \mathbb{E}|u + U|^4 = |u|^4 + 12|u|^2 + 16, \quad (5.7)$$

independence of doublet and singlet signs yields

$$\boxed{\mathbb{E}_A \|\Xi(z + A, V)\|^2 \geq \frac{339}{1000P} |v|^2 + \frac{96}{565PD_R^2} |w|^2, \quad \rho \leq R.} \quad (5.8)$$

At $R = 1/32$,

$$D_R = \frac{193}{32} + \frac{\sqrt{3}}{2} + 10^{-12}, \quad (5.9)$$

and both coefficients in (5.8) exceed $9/10240$. Together with (5.2),

$$\boxed{\mathbb{E}_A \|\Xi(z + A, V)\|^2 + Q_6(z)[V, V] \geq \frac{9}{10240} \|V\|^2.} \quad (5.10)$$

Even if the diagonal comparison is halved, choosing $R = 1/45$ gives the conservative exact corollary

$$\frac{1}{2} \mathbb{E}_A \|\Xi(z + A, V)\|^2 + Q_6(z)[V, V] \geq \frac{1}{2250} \|V\|^2. \quad (5.11)$$

Neither (5.10) nor (5.11) identifies a legal production owner.

6. Exact square-before-average boundary

The response in (3.3) contains $\|\mathbb{E}_f \dot{J}_V\|^2$, whereas (5.3) is $\mathbb{E}_f \|\dot{J}_V\|^2$. Jensen has the wrong direction for the required lower bound. At $z = 0$ and for the symmetric 64-atom law,

$$\Xi(-A, V) = -\Xi(A, V) \quad (6.1)$$

for every V , because the rational denominator is even while all four outputs are odd. Therefore

$$\mathbb{E}_A \Xi(A, V) = 0, \quad Q_6(0)[V, V] = 0, \quad (6.2)$$

although (5.8) is strictly positive. The diagonal-to-mixed promotion is therefore false even after the sextic is appended. A square-before-average realized owner, a matching trace-forest identity, or the separately owned source/low channels are indispensable.

7. Law-free floor-uniformity fails

Set $e = \delta^2$, $u = se_1$, $V = e_1$, and

$$f_\delta(s) = s - \frac{5}{9} \frac{s^3}{s^2 + \delta^2}. \quad (7.1)$$

The only active rows are $2\sqrt{c_0}s$ and $2\sqrt{c_1}f_\delta(s)$. Give s the equal-weight conditional law $\{\delta, 1\}$ and vary both atoms by $s \mapsto s + t$. With the matching trace covariance, the recombined packet is

$$\mathcal{P}^{\text{comp}} = -\frac{1}{2} \text{Var}(C_6(u)V). \quad (7.2)$$

The linear-row Hessian vanishes. Direct differentiation of the rational row gives

$$\boxed{\frac{d^2}{dt^2} \mathcal{P}^{\text{comp}}|_{t=0} = -\frac{5c_1(\delta-1)^2}{324\delta(1+\delta^2)^4} (7\delta^7 + 188\delta^6 + 61\delta^5 + 100\delta^4 + 57\delta^3 + 40\delta^2 + 3\delta + 8).} \quad (7.3)$$

Every displayed polynomial coefficient is positive, and

$$\frac{d^2}{dt^2} \mathcal{P}^{\text{comp}}|_{t=0} = -\frac{10c_1}{81\delta} + O(1). \quad (7.4)$$

Thus current/trace recombination and all-row Fierz cancellation do not imply a law-free floor-uniform response bound. This two-point conditional law has not been realized as the production law, so (7.3) is a method no-go rather than a production counterexample.

8. Standard-Gaussian score transfer repairs the scalar ray

Let X be standard Gaussian and keep the same f_δ . Define

$$P_\delta(t) = -\frac{1}{2} \text{Var}\{C_6((X+t)e_1)e_1\} = -2c_1 \text{Var}\{f_\delta(X+t)\} + \text{constant}. \quad (8.1)$$

The linear-row variance is translation independent. Since $0 < \alpha < 1$,

$$|f_\delta(x)| \leq |x| \quad (8.2)$$

uniformly in $\delta > 0$. Put $m(t) = \mathbb{E}f_\delta(X+t)$, $q(t) = \mathbb{E}f_\delta(X+t)^2$. Gaussian score transfer uses values rather than the singular pointwise second derivative:

$$\begin{aligned} m' &= \mathbb{E}[X f_\delta(X+t)], & m'' &= \mathbb{E}[(X^2 - 1)f_\delta(X+t)], \\ q'' &= \mathbb{E}[(X^2 - 1)f_\delta(X+t)^2], \\ \text{Var}(f_\delta(X+t))'' &= q'' - 2(m')^2 - 2mm''. \end{aligned} \quad (8.3)$$

Using (8.2), Cauchy–Schwarz, $\mathbb{E}X^2 = 1$, $\mathbb{E}X^4 = 3$, and $\mathbb{E}(X^2 - 1)^2 = 2$ gives

$$|\text{Var}(f_\delta(X+t))''| \leq 6 + 2\sqrt{2} + (4 + 2\sqrt{2})t^2. \quad (8.4)$$

Consequently

$$P_\delta''(t) \geq -A - Bt^2, \quad A = 4c_1(3 + \sqrt{2}), \quad B = 4c_1(2 + \sqrt{2}). \quad (8.5)$$

The source Hessian is $9/10$. The once-owned terminal sextic contributes $(9/2)\mathbb{E}(X+t)^4 \geq (9/2)t^4$. Completing the square in $y = t^2$ yields

$$\begin{aligned} \frac{9}{10} + P''_\delta(t) + \frac{9}{2}\mathbb{E}(X+t)^4 &\geq \frac{9}{10} - A - By + \frac{9}{2}y^2 \\ &\geq \boxed{m_G := \frac{9}{10} - A - \frac{B^2}{18}} > \frac{3}{4}. \end{aligned} \quad (8.6)$$

The exact executed value is $m_G = 0.7653207581\dots$. The estimate is uniform in the floor $e = \delta^2$ and uses source and sextic once. It is an actual square-of-mean/current-trace scalar Gaussian packet, not the diagonal comparison of Section 5. It remains one real active ray and contains no spatial shell, chart, low, random-forest, or full A1 owner theorem.

9. What the inherited shell structure supplies

There is real structure beyond a bounded multiplier, but it is presently narrower than the owner-complete mixed response.

- Under the exact R-083 stopped polynomial hypotheses, the six-row stopped-FAR contribution vanishes for collar $C \geq 3$. If a future owner-complete mixed-replica intertwiner preserves those hypotheses, that submodule contributes $C_{\text{far}} = 0$.
- R-087/R-091 give finite support of the rational Cartan principal jet and

$$\mathcal{H}_C(C) \leq 2^{-2\gamma(C-5)} \mathcal{B}_\gamma^{\text{out}}. \quad (9.1)$$

At the proved $\gamma = 7/12$, the square exponent is $7/6$ and the amplitude exponent is $7/12$. This is slower than both the R-131 mixed target exponent 2 and far target exponent 4.

- A new one-use $\gamma = 4$ estimate for the complete mixed rational combination would give amplitude decay $2^{-4(C-5)}$, compatible with the far target. Rewriting it as 2^{-4C} costs the fixed offset 2^{20} . No such $\mathcal{B}_4^{\text{out}}$ theorem is presently proved.

The correct next shell calculation keeps the signed combination

$$JJ - \alpha\tau'JM - \alpha\tau MJ + \alpha^2\tau\tau'MM \quad (9.2)$$

intact, proves the stopped polynomial intertwiner, and computes the leading far paracomposition jets of all τ terms jointly. Termwise commutator cancellation is not an admissible assumption.

10. Proof-search evidence map

The append-only exploration ledger contains the full reusable route records EXP-000496–EXP-000503. This table is only a compact view.

Route	Verdict	Exact boundary or next condition
Paired replica polarization	advanced	Exact representation; common heat and full joint future law are mandatory.
Mixed Pauli–Fierz sum	advanced	Floor is confined to τ, τ' but does not cancel.
Diagonal heat plus sextic	advanced	Global diagonal constants $9/10240$ and $1/2250$; not a mixed response.
Diagonal-to-mixed promotion	failed	Symmetric origin has $\mathbb{E}\Xi = 0$ and $Q_6(0) = 0$.
Law-free floor uniformity	failed	Exact two-point ray diverges as $-10c_1/(81\delta)$.
Gaussian score-transfer ray	advanced	Source–sextic margin $m_G > 3/4$; extend to full production conditional Gaussian owners.
Inherited shell decay	inconclusive	$\gamma = 7/12$ is too slow; prove a complete signed $\gamma = 4$ bound or a stronger cancellation.
Owner synthesis audit	inconclusive	Low, matching energy, shell constants, and the absolute anchor remain missing.

11. Devil’s-advocate review

Objection 1: (5.10) closes the physical response. UPHELD. It does not. Section 6 gives an exact fixture where the diagonal mean-square is positive but the square of the conditional mean and sextic both vanish. The theorem is labelled diagonal throughout.

Objection 2: the two-point divergence disproves production. VALID WITH MITIGATION. It disproves a law-free theorem. The conditional two-point law and its global spatial compatibility are not production authorities. Section 8 shows that standard-Gaussian score transfer has a floor-uniform positive source–sextic scalar-ray margin.

Objection 3: the source has been spent twice. DISMISSED. The diagonal theorem spends no source. The Gaussian-ray theorem uses the $9/10$ source once and the $3/20$ terminal sextic once in the same reduced action Hessian. Neither is added as a second reserve to (2.1).

Objection 4: common heat may be independently replicated. UPHELD. Doing so changes the Hilbert coordinate and the owner. Theorem 3.1 keeps the heat common. Any future implementation which independently replicates it does not instantiate R-132.

Objection 5: the known Cartan tail already gives C_{far} . UPHELD. Its amplitude exponent $7/12$ is too slow, and it covers a narrower module. The missing owner-complete intertwiner and $\gamma = 4$ one-use estimate are stated explicitly.

Objection 6: the scalar Gaussian margin is the absolute anchor. UPHELD. It is a reduced conditional ray with no full low block, chart union, spatial response, or Gibbs normalization. It selects a route but is not the R-127 absolute anchor.

12. Result footer

Result ID: A13-CLASSII-MIXED-REPLICA-GAUSSIAN-RAY-SEXTIC-SHELL-BOUNDARY (R-132).

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.

Precise statement: The complete conditional response has an exact common-heat paired-replica representation and exact mixed Pauli–Fierz formula. A diagonal Rademacher heat--sextic comparison is globally coercive, but an exact

symmetric-origin fixture blocks diagonal-to-mixed promotion. A two-point rational ray rejects law-free floor uniformity, while standard-Gaussian score transfer gives a floor-uniform scalar source--sextic margin above $3/4$.
 Scope: fixed finite cutoff and positive floor; exact conditional algebra; artificial 64-atom diagonal comparison; abstract two-point method no-go; standard-Gaussian one-real-ray positive theorem.
 Dependencies: A1; R-063, R-083, R-087, R-091, R-104, R-123, R-125, R-128, R-129, R-130, and R-131.
 Evidence grade: ANALYTIC, EXACT, EXECUTED.
 Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
 codes/foundations/a13_classii_mixed_replica_gaussian_ray_sextic_shell_boundary_verify.py
 Expected output: R-132 integrated PASS; primary and standard-library independent children PASS; PDF and generated-surface gates PASS.
 Falsification gate: failure of replica polarization; any mixed Fierz contraction; Rademacher moment or compact coefficient; $9/10240$ or $1/2250$ comparison; origin odd cancellation; two-point rational Hessian or $-10/81$ scaled limit; Gaussian score identity, bound, or $m_G > 3/4$; shell-exponent audit; any pinned authority, executable, JSON, exploration, negative, PDF, manifest, or public-surface assertion.
 Tier before / after: T4 / T4; no promotion.
 No-overclaim statement: the diagonal comparison is not the physical mixed response; the two-point law is not a production counterexample; the Gaussian ray is not a full spatial owner or absolute anchor; production constants, both A13 gates, and Sector-A closure remain open.
 Proof complete: false. Sector A closed: false.
 Next required action: prove a production-conditional Gaussian mixed-replica score/small-ball theorem for the complete signed six-row owner, preserving common heat, trace, forest, visits, low, rational and Cartan terms once; prove the stopped polynomial intertwiner and a complete rational $\gamma=4$ far estimate or stronger cancellation; then compute C_{mix} , C_{far} , c_{bal} , D , K , R_0 , S_0 , matching energy, and the shifted-Douglas absolute anchor.

No-overclaim footer. R-132 proves exact replica and mixed-Fierz representations, two diagonal heat-sextic comparison constants, an exact promotion obstruction, a law-free floor no-go, and a floor-uniform positive standard-Gaussian scalar ray. It does not construct or bound the owner-complete production response; prove production C_{mix} , C_{far} , c_{bal} , low constants, matching energy, or an absolute anchor; close either A13 gate; establish $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, or Sector A; or promote any claim to T5–T7.